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Research

Improving routing in wireless sensor networks technology on data aggregation using fused cluster routing algorithm

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Abstract
A crucial challenge in wireless sensors networks (WSNs) is effective communication with tolerable power depreciation of sensor nodes. A bottleneck in the transmission of superfluous data is inevitable in every large network. Energy, latency, and data redundancy are the three identified challenges; to increase existing effectiveness, a trade-off must be made. Existing methods in the literature either employ a distributed or centralized strategy to select a sensor node. The paper proposes a fused clustering-based model with a routing algorithm. In this paper, a dedicated energy-efficient Fused Cluster Routing Algorithm (FCRA) intends to improve data aggregation in WSNs. With applications such as environmental monitoring, catastrophe management, smart agriculture, and urban planning being deployed, a need for low-energy communication protocols breeds more essential. The proposed FCRA integrates hierarchical clustering to optimize routing, thereby reducing transmit redundancy and conserving node energy. This directly contributes to the longevity and reliability of sensor networks in the domains of sustainability. The ideal routing technique for Cluster-Head (CH) identification is dependent on every sensor node's processing capability, and it also assists in handling the intermediaries in the communication path. To enable efficient communication, data from both clusters is gathered utilizing location and temporal correlation, then aggregated. While related to the earlier suggested methods, the proposed method efficacy is shown to be higher, indicating its effectiveness. The efficacy of the proposed algorithm is assessed using modelling criteria such as aggregation, packet transfer percentage, throughput, packet delay, and used energy. This tremendous finding lays emphasis on optimized routing of WSNs which aids in sustainable infrastructures, resilient ecosystems, and informed environmental governance.

Keywords Fused clustering · Data aggregation · Wireless sensor networks · Energy consumption · Correlated routing

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1 Introduction

With the abundant benefits of wireless sensor networks, or WSNs, are now indispensable in almost every control and oversight applications. Among of the significant persistent networks that senses the ambient conditions using a variety of sensing characteristics is the WSN. In order to communicate through other devices, the WSN is encompassed with numerous sensor nodes, encompassing base stations, sink nodes, as well as source nodes. Numerous scarce in resources sensor nodes (SNs) comprise WSNs. The purpose of the SNs is to perceive their external location and send sensory information to their base station (BS) [1]. Attributable to its factors in maintaining domains such as military surveillance, nursing care or abetted living, measuring energy of at all phenomenon, real-time monitoring in commercial processes, critical infrastructures such as environment monitoring and natural disaster [2], WSNs are by far the most flourishing area to investigate. The effectiveness with the above technologies is hampered by a number of factors.

Wireless Sensor Networks (WSNs) have intrinsic challenges affecting the efficiency, endurance, and performance of networks. The major aspect is the characteristics of WSNs that consist of battery-powered sensor nodes deployed into distant or unreachable positions where less battery life spans, inefficient communication, excess in data transfer, congestion, and collision occur. All these issues together require some combination of energy-efficient protocols, robust data aggregation methods, latency-aware communication strategies, and other advanced security mechanisms. Optimization of these parameters is vital for making WSNs reliable and efficient in actual applications. In resource-constrained WSNs, data aggregation stands as a skilled mechanism that preserves energy, and this has been explored quite well in the existing works. In-network data aggregations are an essential part in keeping energy-saving by removal of redundant and non-essential information. Wireless Sensor Networks (WSNs) are subjected to some intrinsic challenges impinging on their efficiency, longevity, and performance. Primarily, WSNs are usually made up of sensor nodes that are usually battery-operated and deployed in locations that can be remote or inaccessible, leading to reduced life-span of batteries, inefficient communication, excessive data transmission, congestion, and collisions. All these together bring the need for some combinations of energy-efficient protocols, robust data aggregation methods, latency-aware communications, and various advanced security mechanisms. These factors' optimization is very important in making WSNs reliable and efficient in real-world applications. Within resource-constrained WSNs, aggregation of data stands as a competent means of conserving energy and is extensively researched in existing works. By eliminating duplicate and unessential information, in-network data aggregation stands out as an important aspect of energy saving. Wireless sensor networks are an integral part of the digital infrastructure of smart and sustainable systems. These applications span across a wide range of sectors which are vital to achieving the United Nations Sustainable Development Goals (SDGs) on climate action, clean water monitoring, precision agriculture, and sustainable urban development. However, energy being a perennial challenge for wireless sensor networks (WSNs) puts a limitation on operational longevity and environmental footprint. As a result, efficient data aggregation and routing are indispensable not only in a technical context but also from a sustainability perspective.

Following are the key energy-consuming operations in WSNs-data receiving, data forwarding, data processing, and transmission. These energy-consuming activities are typically governed by certain factors, such as the distance a sensor node may be required to transmit, defective transmitters, limited battery power, waiting periods for consuming energy, waiting periods for receiving packets, acknowledgements, etc. Most of the existing technologies, traces, innovative methods, et cetera are less attractive and have never been used in energy recovery, when fixing the power source for the WSN, especially when data transmission becomes a great advantage. Conservation of energy boosts the life expectancy of WSNs, yet, on the other hand, corresponding technological conservation techniques might present fly in the ointment; the increase of energy management becomes a hot point of research.

The densely placed sensor nodes often give rise to redundant data in wireless sensor networks since several nodes monitor the same event or area. Although redundancy can give better reliability and fault tolerance of data, redundant data produced in excess leads to increased energy consumption, bandwidth wasted, and unwanted processing overhead. Data aggregation and compression techniques are necessary so that the nodes may process and filter the information before transmission. However, to use these techniques, there exist efficient algorithms requiring a trade-off between data accuracy and energy efficiency. If not carefully managed by the applications, data aggregation might lead to imprecise value judgments due to errors or omitting critical details. An example scenario would be: a dense placement of sensor nodes in an area or around an event usually results in redundant data generation in wireless sensor networks. Redundancy, however, can ensure the reliability and fault tolerance of data being

transmitted. On the contrary, excessive redundant data are likely to consume energy, waste bandwidth, and create burdens because of needless processing overhead. Data aggregation and compression techniques are necessary so that the nodes may process and filter the information before transmission. However, to use such techniques, there exist efficient algorithms requiring a trade-off between data accuracy and energy efficiency. Otherwise, data aggregation might lead to imprecise value judgments due to errors or omitting critical details.

The limitations posed by energy resource constraints, redundancy of data, and latency in communication present major barriers to the successful adoption of WSNs. Solutions to these major challenges must identify newer techniques for improving energy efficiency, optimizing data transmission, and reducing latency. Future enhancements should entail work on adaptive and intelligent schemes that can be designed and implemented to overcome limitations. Continued importance of WSNs in an array of areas with need for real-time and reliable data collection in the future, in overcoming such challenges, must be ensured through further investigations. Revolutionising the challenges limited energy resources, data redundancies and communication latencies suggest deployment issues concerning WSNs. Novel solutions or innovative techniques to improve energy, optimize data transmission, and reduce delays are required. Future lines of research must investigate adaptive and intelligent strategies that would be developed and implemented to overcome these limitations. According to these new challenges, future development should be done on WSNs, which would prove necessary in almost all sectors requiring real-time data collection. There are restrictions on available energy resources, redundancies of data, and latencies of communications. These challenges pose major hindrance to proper implementations of WSNs. Solutions to these central problems would include newer techniques improving energy efficiency, optimizing data transmission, and reducing latencies. Future research must focus on creating adaptive and intelligent methods to overcome these limitations.

The key constraints for all sorts of WSN applications are restricted energy resources besides network interval. When harvesting energy is not used, energy savings can stand achieved through putting the nodes in energy-saving mode and activating them whenever anticipated. In hydrocarbon networks, there really are two basic objectives to attain. The primary goal is to change data from the detecting node to the drain using as little energy as possible, and the second goal is to relay events using a well-organized routing system to reduce delays. Despite prudent use of power, WSN shall ensure scalability and the capacity to manage fatigued nodes. Both heterogeneity and mobility must be managed in a large network. In principle, cross layer design is used to order to effectively achieve, energy efficiency, and data rate [2–4]. The behaviour of effective CH range amongst the nodes is widely discussed in classic clustering approaches for homogeneous or heterogeneous WSN. As in Low Energy Adaptive Clustering Hierarchical routing protocol (LEACH), CH is determined based on characteristics such as location, node density, and uniform probability among nodes. Sink-generated data gathering from the cluster-heads at predetermined intervals is advantageous in only certain applications necessitating continual monitoring. Conventional systems may not be flexible in such situations, necessitating the development of a new method. Selection of multihopping is depicted in Fig. 1. The routing duties include unicast, multicast, broadcast, single-hop, multi-hop, and mixed routing, to name a few. Various challenges confronted in the clustering pitch with routing include grade retention, data reporting, and conglomeration of node/link, fault tolerance, then network dynamics.

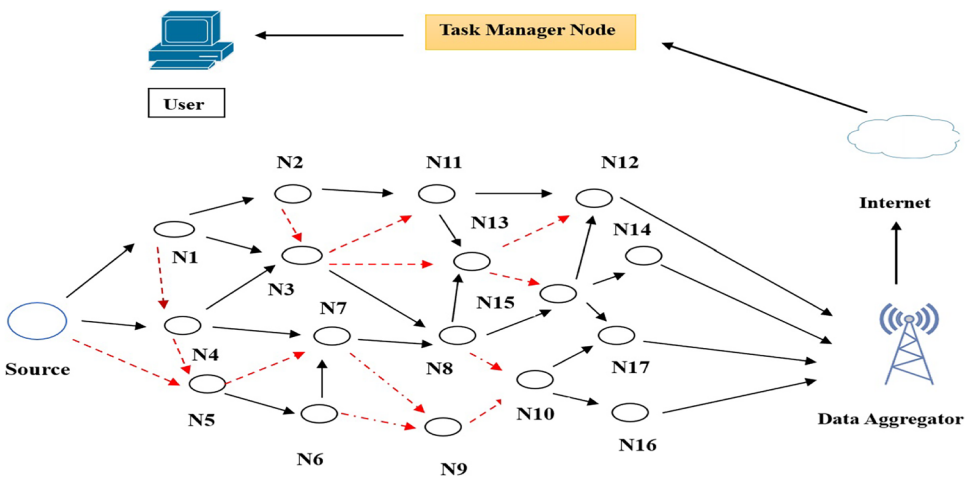


Fig. 1 Selection of Multihopping with mixed routing

The whole system region is divided into discrete clusters by the current energy-saving techniques. The CH is in responsibility of each group [5]. A CH's primary responsibility is to collect sensed information from its members of the cluster (CMs) and communicate it to the base station (BS), therefore significantly extending the networks' lifespan. On the other hand, the SNs closer to the CH are the primary reason of the employed clustering approaches' disadvantage. Compared to the faraway SNs, those SNs send more packets. They are therefore at risk of perished prematurely. There are two potential approaches to address this issue: first, divide the area being observed region into multiple distinct segments of the network that might not be capable to connect to the BS, which would result in subpar network performance [6]. Furthermore, in order to divide the energy usage across the CMs, certain clustering techniques activate the CH its part on every sensor. However, the present methods require a lot of overhead messages as well as which slows down transmission thus degrades network performance. Substantial effort has been used to solve difficulties with routing using the techniques that were described.

WSNs, which are made up of several inexpensive sensor nodes widely distributed in various areas, are often employed for monitoring and controlling particular environments. Sensor networks have piqued the interest scientific community as a way to collect information about the environment. Yet, it is difficult to recharge or replace nodes because they are frequently placed in hostile and inaccessible regions and are fuelled by batteries with a limited capacity. Because of this, managing energy in WSNs is a crucial issue [7]. One of the best strategies for successfully managing energy in these networks is clustering. In order to minimize consumption of energy, throughout its entirety latency, and superfluous network disputes, it is imperative to ensure that packet priority among sensor nodes be taken into account throughout the duty cycle processing. Data aggregation is a powerful strategy for conserving energy in resource-constrained WSNs, and it has been extensively examined in previous research. In-network data aggregation is critical for reducing energy consumption by removing redundant and superfluous data. It separates and sends only the content that you want to the sink. Both the time and the cost of communication are reduced as a result. The cable network density and the nature of physical phenomena engage in a crucial part in the presentation of reality with both spatial and temporal domain correlation. Aside from the WSN's collaborative nature, spatial and temporal correlations yield significant potential returns in the establishment of competent security mechanisms. To eliminate duplication information, just processing options are picked for sending information rather than getting information from all nodes [8, 9].

The proposed method employs a Fused-clustering based routing algorithm (FCRA) for data aggregation and routing is unique in that it incorporates a node's managing capacity into cluster creation for reliable data forwarding. This would optimize the aggregation of data while reducing energy consumption in WSNs. With an emphasis on energy efficiency, the proposed method will contribute to the establishment of long-term, self-sustaining sensor networks in some of the most critical application areas in sustainable development. The primary objectives are: During aggregation, temporal and spatial correlations in data are engaged into consideration, resulting in energy preservation and low bandwidth. Each sensor node in a WSN has different skills to deal with a group of nodes throughout clustering along with data transmission. This is referred to as a node's capability of managing a node. The primary elements of clustering are: Clusters are generated based on network distance, and CH assortment is used for appropriate data gathered. Communication of spatiotemporal data collected from cluster head to base station for efficient energy use. Develop a multi-hop path routing approach based on different clustering and guidelines are followed to extend the cable network existence. In this paper, the relevance of a node to its node management capability is discussed, as well as its residual energy. The electricity demand of nodes is managed in the proposed model for data aggregation and routing algorithm by minimising redundant information. The structure of the paper is as follows: A comprehensive survey is given in Sect. 2, Sect. 3 elaborates on the Methodology followed by experimental arrangement and outcomes in Sect. 4. At last, Sect. 5 concludes the work with future improvements.

2 Review of the existing literature

Numerous studies offered novel methods to spread the life of the sensor networks. In optimization schemes alike LEACH and its derivatives Efficient Cluster-Based Self-Organization Algorithm (ECSOA), Stable Election Protocol (SEP), Energy Efficient Clustering Protocol for Underwater Wireless Sensor Networks (ECPUWSN), Power-Efficient Gathering in Sensor Information System (PEGASIS) and Sparsity-Aware Energy Efficient Clustering Protocol for Underwater Wireless Sensor Networks (SEEC) [10–13], heuristics were a typical way to finding Cluster-Head from a group of nodes in a bunch. A

homogeneous CH allocation wasn't really accomplished because the clustering algorithm was predetermined in the above processes. These methods ensured optimal energy savings at the local level, but not at the global level [14, 15]. A lesser number of nodes with network homogeneity were anticipated in LEACH. Thus, assures consistent frequency, but because single hop transmission consumes a huge amount of energy, this is inappropriate for big-scale WSNs. CH is usually chosen, which is based on the intensity of the conventional signal and the number of steps. Previously, Euclidean distance amongst any nodes to the Cluster-Head was commonly used in certain existing methods. That resulted in non-uniform node density across group of clusters. If CH is chosen based on the frequency strength, the likelihood of a component fetching Cluster-Head several times is considerable, resulting in an energy hole problem [16, 17].

Algorithms like Energy-Efficient Data Aggregation and Transfer (EDT), Distributed algorithm for Integrated tree Construction and data Aggregation (DAITCADA), Data Routing for In-Network Aggregation (DRINA), and Cluster-Based Data Aggregation (CDA) are largely concerned with clustering and directing in multi-level heterogeneous environments. That employed a global node scheduling method that eliminated collisions., all of the sensor nodes' information gathered was sent to the access point in the shortest amount of time possible in DAITCADA. The accumulation tree was built utilising a structure of nodes, with each person having a control message of equal, smaller, or more than the base station. Using clustering and data accretion, the duplicate data transported from the cluster member (CM) to the CH while from the heading to the core network was reduced in order to save energy.

In Cluster-Based Data Aggregation for identifying the ideal way, energy conscious route method is utilised to limit the use of the energy, latency, and bandwidth. Data aggregation in EDT was divided into two phases: aggregation and adaption. When redundant raw data from a certain time period was eliminated during the aggregation process, correlations between the data were found. A joining function involving sets-similarity was employed in the adaption phase to find redundant data and gradually transmit it to the CH. Optimization methods including suffix filtering as well as the k-means algorithm were applied to lower data latency [18, 19]. In order to minimize the number of communications utilized exclusively for tree-arranged routing, optimize the paths overlapping counts, speed up aggregation, enhance data clustering, and publish guidelines, DRINA created a novel in-network amalgamation with routing of data. However, because the next node was chosen based on destination node, there was even a longer transmission time and delay. In many other studies, only data aggregation was considered, but data that was spatially and temporally connected was just not [9, 20]. A DE-LEACH (Distance and Energy Aware LEACH) proposed a distance and energy conscious LEACH. The solution used adaptive CH selection, which resulted in a significantly lower packet drop ratio. Nevertheless, this approach had the drawback of departing accumulated data in the event that Cluster-Head failed, and the previously mentioned drawback had been brought about by the implementation of an accretion process. A time series approximation method with single-hop segmentation has been developed to reduce the transmission of data. The technique has been compared to other techniques, such as the ARMA (Autoregressive Moving Average model) as well as normalized LMS (Least Mean Square) models, that eventually reduced a wireless transmission of data, in order to calculate regional sampling felt. Because of a link between the sensing data, the data aggregation is employed to minimize no fewer than two packets to one. Data aggregation leads to a greater improvement in the total lifespan of the network by lowering the total quantity of packets utilized for communications, as an enormous quantity of power was used for data transfer and reception [21]. CH is chosen only on the basis of communication between CHs and Cluster-Messages, not even from the standpoint of the sink. The Euclidean distance or the power consumed of the sensor network are used to define routing protocols. Any of the above-mentioned routing protocols ignores Crossing Over Distance (COD). Despite the fact that just a few protocols addressed delay, the majority of the protocols did not examine any of the other performance indicators [22]. Furthermore, while mean average threshold values are taken into interpretation for data aggregation and scheduling, on the other hand spatial–temporal data were not taken into account in most grouping and communication overhead.

A clustering protocol using genetic algorithm was proposed [23]. This works with the fitness function rising the lifespan of the network. A congestion aware routing mechanism is proposed employing fuzzy logic and greedy approach [24]. They are capable to fine-tune the transmission rate. A different approach is proposed in choosing the CH, founded on the maximum score [25]. Energy consumption and improved delivery rates were achieved. A two-level strategy deploying clustering grounded on fuzzy logic mechanism and content-based routing mechanism was proposed to improve the routing performance achieving improved lifetime, increase in quantity of active nodes and energy consumed with packet delivery rate [26]. The modern technologies: big data and internet of things proved to progress the performance of the network. In addition to this, with the integration of fog computing an overlapping clustering-based method is

proposed [27]. With the incorporation of fog, cloud and internet of things, the response time, transfer rate and lifetime are proved to be increased along with energy consumption [28].

Most significant issues in WSNs include consumption of energy, network endurance, and accessibility. The research presented here suggests a mechanism for dividing network region into square cells in order on a topographical basis. The node with the most energy is chosen as an aggregate in each of the cells [29]. A robust approach to handle traffic in wireless sensor networks is proposed [30]. An adaptable mechanism which can extract features with dynamic adjustments routing based on applications were provided. A hybrid approach for routing technique was proposed using threshold-based information gathering in heterogeneous environment with the amalgamation of internet of things is proposed to improve load balance [31]. An approach to improve energy consumption and to extend the lifespan is proposed using LEACH protocol [32]. To improve the performance of WSNs, a methodology integrating artificial bee colony approach and ant colony optimization is proposed [33]. The performance is proved to be efficient in rapports of energy consumed, improved lifeline, and data packet transfer rate.

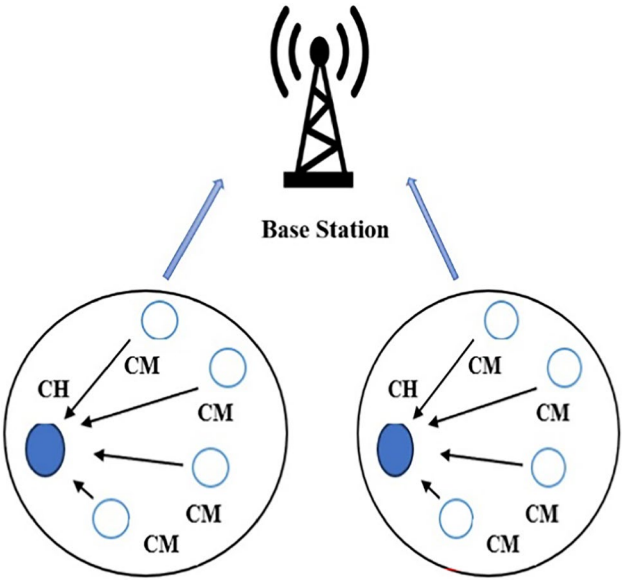
Existing techniques for determining the CH include Distributed Energy Efficient Clustering (DEEC) and a Hybrid Energy-Efficient Distributed Clustering (HEED), which are calculating diverse nodes' utilization of energy utilizing weight info and residual energy, accordingly. To reduce bandwidth and energy usage, an aggregation technique using dynamic programming has been proposed [34]. To reduce communication and conserve bandwidth along with energy, the data accumulation process is used at the clustering level. Improving WSNs' energy economy and network resilience is the main goal of this investigation [35]. To be able to improve the efficacy of energy and optimize the stability of the network of the WSN, this work suggests an approach in which hybrid particle swarm optimization is coupled with improved low-energy adaptive clustering hierarchy (HPSO-ILEACH) to obtain choosing cluster head in cases of data aggregate. Using the above method, HPSO computes the cluster head, the separation between each node in the cluster, and the nodes' residual energy. A rapid approach of aggregating dynamic WSN data that utilizes micro-cluster adaptive learning has been proposed in order to enhance the calibre of data aggregation [36]. The Kalman filter is presented and the network of wireless sensors data is analysed under the evolutionary acquiring micro-cluster. An intelligent distributed technique based on ant colony approach is presented in this paper [36]. It targets improving energy efficiency. With an emphasis on improving the lifetime of the network and its capability in internet of things-based sensor networks utilizing a Distributed Clustering Mechanism (DCM) along with an effective routing technique, the Teacher Learner Firefly Optimizer (TLFFO), this research presents a novel strategy to overcome these barriers [36]. Furthermore, many existing intrusion detection methods do not take into account load balancing with grouping for lowering energy dissipation when executing security algorithms. In a review of relevant literature, it is concluded that, aside from data aggregation, numerous protocols using routing and classification methods helps organization endurance. Numerous clustering methods have been created with the goal of stabilising CH and spreading other locations that use that CH. CH is chosen only on the basis of communication between CHs and Cluster-Messages, not even from the standpoint of the sink. The Euclidean distance or the power consumption of the sensor network are used to define routing protocols. Thus, there is a need to develop a mechanism which can reduce energy usage and minimal delay. The paper intends to develop an energy-efficient Fused Cluster Routing Algorithm (FCRA) and evaluate it for data aggregation in Wireless Sensor Networks, while enhancing lifetime and conserving energy in the process. This research will ultimately help the greater sustainable development agenda with robust, low-power sensing options for environmental and societal monitoring applications. The proposed approach lies in the strategy of managing the capacity of nodes with the structuring of clusters for improved and reliable data routing. Also, selecting cluster head is proficient in data aggregation. Targeting multi-path routing with improved lifespan of the network.

3 Methodology

3.1 Design

Consider the case with a number of nodes of N, with a considerable of 'r' transmission, and an initial energy of 'e' in a centralized network with randomized node deployment. Straight mainly transmitted only if the mobile networks are inside the sink's coverage range. While connecting to both the sink, and the source nodes employs intermediary nodes as relays, resulting in numerous hops. The topology of the architecture is represented in Fig. 2.

Fig. 2 Network topology



3.2 Energy optimization scheme

The variants with the effects used in the proposed method varies with the distributed adaptive clustering protocol in rapports of cross-over distance. In (Eq. 1) and (Eq. 2), the resources expended conveying '*b*' bits of information across a distance '*dt*' is stated as

$$Ener(b,d) = \left(b * E_{energy} + \rho_{frse} * dt^2\right) ifdt < dt_{crossover} \tag{1}$$

$$Ener(b,d) = \left(b * E_{ener} + \rho_{mpfg} * dt^4\right) ifdt > dt_{crossover} \tag{2}$$

The energy consumed for transmission shall be figured with the above Eqs. 1 and 2, whereas the $dt_{crossover}$ is arrived by (vertex range/2). In turn $Ener(b,d)$ shall be computed with the quantity of energy consumed in the space available (*frse*) and dynamic multi-path propagation fading (*mpfg*).

3.3 Proposed algorithm (FCRA)

Dependent on the rate clustering and routing approach incorporates two primary methodologies: each sensors single node capability of Node Management (CNM) and cross-over length. These abovementioned strategies have never been used in any study related to clustering and directing of WSNs in the literature review. Additionally, it is predicted that resource consumption of particular nodes can indeed be prevented by utilising node controlling capacity to make sure a constant node density over all clusters. Because the node penetration diminishes with time, using Euclidean distance is also not a good idea [37]. As a result, the cross-over length is taken into account when developing modelling techniques in order to attain a high packet delivery percentage in the proposed work. Operational framework, node localisation, cluster creation, CH selection, clustered upgradation, and transmitting data are the components of the suggested method. The position update information is depicted in Algorithm 1 whereby sensor nodes receive broadcast messages from the source component of the system. The source network node network id, identifier, and position are modified by revisiting all non-visited components.

Algorithm 1 Nodes transmit information about their location to both the drain

Input: Message (pr_id, id_no, pr_a, q_b, visited-node_list1)
Process: Visit every node that hasn't been visited to determine the message's parameters.
Output: Location of the node (pr_a and pr_b)
begin
{
msg (pr_id, id_no, pr_a, q_b, visited-node_list1)
for each i Є k do
{
$_{praddr}(i) \longleftarrow$ index (o)
for each j Є k do
{
if (node1_visited_list1$[v]$==index (o)
del (msg)
else
pr_id(i) $\longleftarrow$ $pr_{addr}(o)$
id_no(i) $\longleftarrow$broadcast_id(o)
} end - if
} end the for loop
} end the for loop
} end

3.4 Development of cluster and finding a CH

The collecting of data from either the sensor network is the first step in the clustering process. The drain begins the apply that knowledge process in a centralised manner after gathering the data. Afterwards when, cluster maintenance, such as CH re-election, is spread across the sensors. The sink routinely pulls cluster maintenance updates in order to maintain better communication between the sensor network as well as the drain. After learning the location coordinates of sensor network, the data packet accepts the CH with the highest probability first. The data packet uses node information to compute the node's likelihood of fetching a CH, which is then recorded in the node's likelihood table. As during development process, a diminishing order possibility table is used, with both the CH being the node with the highest probability. This freshly formed CH's information was broadcast to all nodes, and the CH begins getting data. Clustering is a very popular technique in WSNs, wherein it works to improve network scalability, energy efficiency, and data aggregation. In a clustered WSN, nodes are grouped into clusters, each of which is managed by a designated Cluster Head (CH). The CH collects the data from member nodes, aggregates it, and then sends this data to the base station. Thus, through clustering, energy consumption can be minimized since most individual nodes do not communicate directly with the base station. Moreover, the performance of the network improves. On the other hand, the CH has to be always on, which leads to fast draining of the energy in certain nodes. To cope with this, CH rotation is used, where the CH role is shared among different nodes-stated periodically within the cluster. This will then balance the load of energy cost across the entire network and prevent early node failure while adding

years into the lifetime of the entire network. Such CH rotation strategies include the likes of probabilistic selection, round-robin rotation, and energy-aware selection, all of which provide a balanced consumption of energy and communication overhead, as well as network stability. Such increased performance would yield lifetime benefits in the network, but this also increases difficulty with synchronization overheads and re-clustering delays. As a synchronous system with adaptive formation schemes, CH transitions should be very smooth. There are 19 nodes considered in choosing a CH. The CH is chosen as the very first node with both the maximum probability value, including one neighbours as its CMs. As both a result, node 12 is designated as the CH, with neighbours 7, 10, 13, and 14 serving as the CMs. Those nodes are then deleted from either the probability table, and also the process is repeated once all of the nodes fall underneath one of the clusters. There are two lists, one for CH and one for CM. Every other node is taken offline and placed in the appropriate list [38]. Overall category of a node is determined by whether it belongs to the CH or CM lists.

Algorithm 2 Forward path method

In: Drain discrimination message, drain _discmsg

Process: New packet is forwarded

Out: r_p is used to choose the routing path.

```
begin
{
 drain_dismsg(msg_type, id_no, v_pr, hopcount, α)
  {
   while (a≤b)
   {
    if (v_pr == ID(j))
        del (pkt)
    else if(pkt==old)
       del (pkt)
    else if(pkt==new)
       while (The packet fails to arrive at the aggregator)
           forwarder selection
     end while
    end if
    end if
   end if
  }
   else
      Transmit the packet again
   a ⟵ a + 1
 } end while

} end
```

Regardless of the overall volume of control packets being reduced, there is an excessive amount of data that inundates the data packet. Another benefit of the proposed approach is that it moves the accumulation job at hand to the CHs, thus avoiding this issue. In these cases, CHs use recent readings and aggregated values so rather than raw data to find the cause-and-effect relationship region. Reduced overhead results in parallel computing, and data processing reduces the operating costs.

4 Results and discussion

4.1 Experimental setup and evaluation

In the context of our proposed method, our work has been implemented as part of an implementation of the Network Simulator 3 (NS3) software tool that is running in the Ubuntu based operating system. Because NS3 is a simulation tool, we believe it is better suited for simulating the number of different communication networks and thus recommend it for testing sleep work schedules scenarios in WSN environments. CDA, DAITCADA, EDT and DRINA were compared to find enhancements in the existing algorithms, and NS-2 simulator results were used to do this. Both data directing in network accumulation and effective data gathering attentive of spatio-temporal associated techniques possess many of the same basic characteristics. Nodes having the same transmission radius are indiscriminately arranged in the replication extent of 400×400 m. In order to randomly generate the network topology, pick dissimilar set of linkage nodes. The following simulation parameters are provided by default, but you may change these parameters built on the assessment conditions. The modelling of energy and broadcast methods are examined and the parameters and corresponding values are depicted in the Table 1.

4.2 Average packet proportion (APP)

The proportion of completed packet transmissions to total packet transmissions per unit time. By adjusting the detecting intervals and period, FCRA (Fused Clustering based routing algorithm) provides a similar aggregation rate (AR) when compared with conventional schemes such as DRINA, CBA, EDT. Correlation existed of similar ideals (not included in DRINA), clustering radius, and transmission cost modification, FCRA delivers fewer packets than other techniques. Only the essential information should be kept obtained from data correlation from both the spacetime domain, deleting duplicate packets to a larger extent in FCRA than in Cluster-Based Data Aggregation in WSNs (CBDAW) and Data Collection Aware of Spatio-Temporal Correlation (DCSTC). As a result, an effective interventions ratio is attained, and network capacity is increased by lowering energy usage. The graph illustrating the aggregation

Table 1 List of parameters

Parameters	Values
Simulation zone	400 * 400 m
System magnitude	90
Initial energy	140 J
Cluster radius	75 m
Period of packet	0.3–0.9 s
Bandwidth	800 Kbps
Channel category	Multipath
Channel freq. rate	2.6 GHz
Determined nodes	500
W1 (Weight factor)	0.6
W2 (Weight factor)	0.9
Transmit capacity	0.04 J
Receive capacity	0.03 J
Size of the message	6000 bits

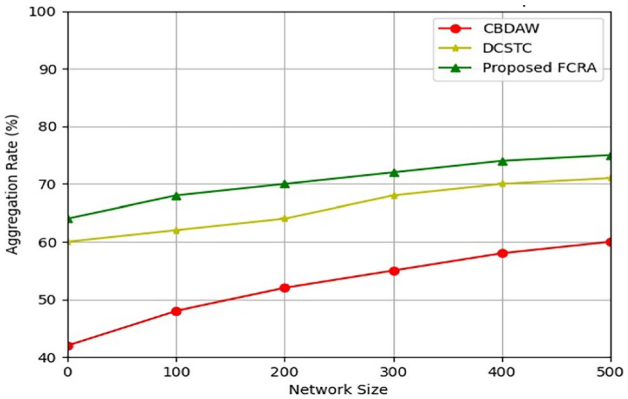


Fig. 3 Aggregation ratio in relation to packet size of the network

ratio in relation to packet size of the network is depicted in Fig. 3. According illustrated in the aggregation ratio is determined for a message size of 5500 bits with a node frequency ranging from 60 to 350.

4.3 Delay

Accordingly, delay is the time it consumes for a transmission to transit from transmitter to the receiver all over a network. The quantity of bits transported is reduced dramatically in FCRA because only the consolidated packets are distributed. However, control messages are employed in Data Routing for In-Network Aggregation to create a routing architecture, which increases the packet count and thus the delay. Latency continues to rise as the number of time slots increases, which has a direct influence on the volume of nodes. However, with FCRA, the correlation section as well as fault tolerance are additionally considered in forwarder node. As a result, the quantity of packets sent lowers, lowering the transmission latency in FCRA. In comparison to Cluster-Based Data Aggregation in WSNs (CBDAW) and DCSTC, data latency in FCRA is shown to be reduced leading to a larger aggregation rate. There is an increase in data latency during certain time intervals, it is depicted in Fig. 4.

4.4 Throughput

CBDAW as well as DCSTC both have their throughput reduced with an increase in the packet loss probability. This method uses multi-hop communication with no disjoint regions for congestion, leading to a higher congestion level and thus a lower throughput level. This resulted in considerable congestion prevailing within the inter-cluster communications alone, bringing about packet loss and reduced throughput. Limiting the duration of the time slots increases the chances of packet loss, more so from nodes with more antecedents. Figure 5 shows the throughput across various time slots where the performance of the proposed FCRA is seen to be better than that of other approaches in improving throughput.

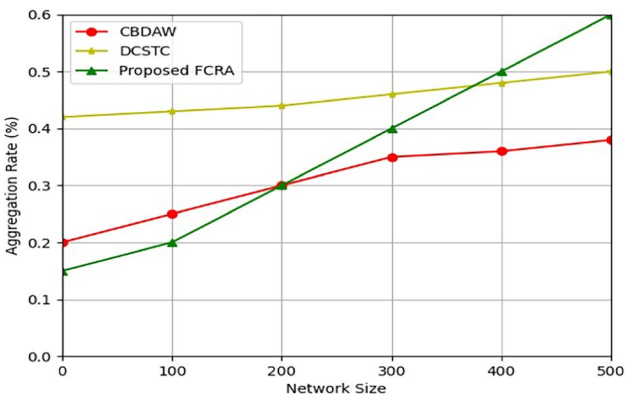


Fig. 4 Increase in latency with time slots

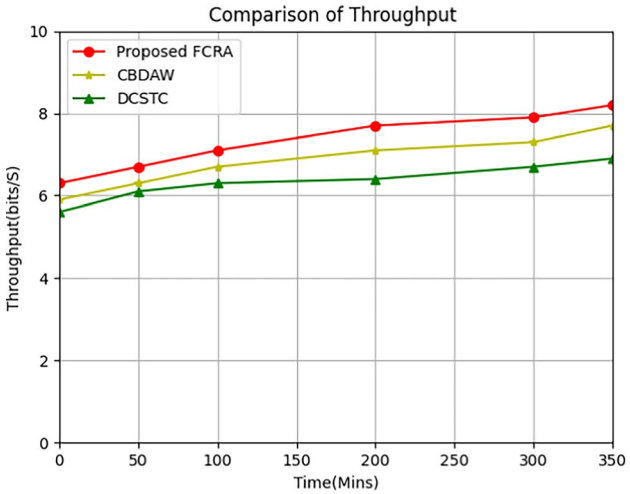


Fig. 5 Throughput comparison

The outcome contains significant implications for sustainable development. More importantly, by increasing the possibility of longer-lasting and more efficient operations of wireless sensor networks, the proposed model minimizes electronic waste, and minimizes battery replacement frequency. Thus, it is appropriate for long-term usages such as environmental caring, intelligent agriculture, and infrastructure watching-all of which are closely related to sustainability goals. Since the algorithm is adaptable, it can work in various environmental constraints, which makes it a very resilient and eco-friendly solution for future technologies.

5 Conclusion

FCRA, which combines temporal and spatial connection of data required to implement clustering and routing algorithms. In this way, clusters have nodes that retain more energy staying in the CH and nodes that really are one hop away from that node join the CM. It is suggested which keeping uniform node density among all the clusters can be achieved by applying the cluster node handling capacity. With regard to these various parameters, the proposed method outperforms cluster-based data aggregation in WSNs and data collection aware of spatio-temporal correlation. Also, the proposed methods contribute to the sustainable deployment of wireless sensor networks. If there were more sensor nodes, this same situation might be more visible. Experiments may therefore focus on employing a larger number of devices in the future. While the aforementioned challenges nevertheless require further investigation, implementation issues, network security, as well as data quality with numerical evaluation shall remain a topic for further research. Regarding packet latency, aggregation rate, routing overhead, throughput and energy consumption factors, the suggested model performs better.

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Declarations

Conflicts of interest The authors declare no competing interests.

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Clinical trial Not applicable.

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